

Group 18 - Database Description

Decision of the Database Management System (DBMS): We use PostgreSQL 18 because it is an open-source relational DBMS suitable for server-side e-commerce applications. It supports relational modelling, primary keys, foreign keys, check constraints, enum types, indexing, JSONB data, transaction-safe operations, etc. PostgreSQL 18 is really appropriate for this project, where product, order, invoice, payment, refund, and user-role data must be stored consistently and queried reliably.

users

#	PK	FK	Column name	Data type	Mandatory	Description
1	x		user_id	UUID	Yes	Auto-generated unique identifier for each user account
2			username	VARCHAR(100)	Yes	Login username used by administrators or product managers. Must be unique across the system
3			email	VARCHAR(255)	Yes	User email address. Must be unique. Used for login and notifications.
4			password_hash	TEXT	Yes	Bcrypt-hashed password. Plain text password is never stored
5			status	user_status_enum	Yes	Account status: ACTIVE, DEACTIVATED, BLOCKED. Default: ACTIVE.

Constraints:

- PRIMARY KEY (user_id)
- UNIQUE (username)

- UNIQUE (email)
- status must be one of ACTIVE, DEACTIVATED, BLOCKED

roles

#	PK	FK	Column name	Data type	Mandatory	Description
1	x		role_id	UUID	Yes	Auto-generated unique identifier for a role
2			role_name	VARCHAR(100)	Yes	Role name. Must be unique. Examples: ADMIN, PRODUCT_MANAGER.
3			description	TEXT	No	Description of the role and its responsibility

Constraints:

- PRIMARY KEY (role_id)
- UNIQUE (role_name)

user_roles

#	PK	FK	Column name	Data type	Mandatory	Description
1	x	x	user_id	UUID	Yes	References the user assigned to a role: users.user_id. Part of composite primary key
2	x	x	role_id	UUID	Yes	Reference to roles.role_id. Part of composite primary key

Constraints:

- PRIMARY KEY (user_id, role_id)
- FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE
- FOREIGN KEY (role_id) REFERENCES roles(role_id) ON DELETE CASCADE
- This table implements the many-to-many relationship between users and roles

products

#	PK	FK	Column name	Data type	Mandatory	Description
1	x		product_id	UUID	Yes	Unique identifier of a product.
2			product_type	product_type_enum	Yes	Type of product: BOOK, CD, DVD, or NEWSPAPER.
3			title	VARCHAR(255)	Yes	Product title
4			category	VARCHAR(100)	Yes	Product category
5			general_description	TEXT	No	General product description, such as condition or return condition.
6			height	NUMERIC(10,2)	Yes	Product height
7			width	NUMERIC(10,2)	Yes	Product width
8			length	NUMERIC(10,2)	Yes	Product length
9			weight	NUMERIC(10,2)	Yes	Product weight
10			barcode	VARCHAR(100)	Yes	Product barcode

11			original_value	NUMERIC(14,2)	Yes	Original product value before VAT
12			current_price	NUMERIC(14,2)	Yes	Current product price before VAT
13			stock_quantity	INTEGER	Yes	Available quantity in stock
14			status	product_status_enum	Yes	Product status: ACTIVE, DEACTIVATED, or DELETED.
15			created_at	TIMESTAMP	Yes	Product creation timestamp
16			updated_at	TIMESTAMP	Yes	Last product update timestamp

Constraints:

- PRIMARY KEY (product_id)
- UNIQUE (barcode)
- CHECK (height >= 0 AND width >= 0 AND length >= 0 AND weight >= 0)
- CHECK (original_value >= 0)
- CHECK (current_price >= 0)
- CHECK (current_price >= original_value * 0.30)
- CHECK (current_price <= original_value * 1.50)
- CHECK (stock_quantity >= 0)
- CHECK (status <> 'DELETED' OR stock_quantity = 0)
- updated_at is automatically maintained by trigger trg_products_updated_at

books

#	PK	FK	Column name	Data type	Mandatory	Description
1	x	x	product_id	UUID	Yes	ID of the corresponding product.
2			authors	JSONB	Yes	List of book authors.
3			cover_type	VARCHAR(50)	Yes	Book cover type: PAPERBACK or HARDCOVER.
4			publisher	VARCHAR(255)	Yes	Book publisher.
5			publication_date	DATE	Yes	Book publication date.
6			number_of_pages	INTEGER	No	Number of pages.
7			language	VARCHAR(100)	No	Book language.
8			genre	VARCHAR(100)	No	Book genre.

Constraints:

- PRIMARY KEY (product_id)
- FOREIGN KEY (product_id) REFERENCES products(product_id) ON DELETE CASCADE
- CHECK (number_of_pages IS NULL OR number_of_pages > 0)
- CHECK (cover_type IN ('PAPERBACK', 'HARDCOVER'))
- authors is stored as JSONB because authors are treated as product attributes in the final ERD rather than separate entities.

cds

#	PK	FK	Column name	Data type	Mandatory	Description
1	x	x	product_id	UUID	Yes	ID of the corresponding product.
2			artists	JSONB	Yes	List of CD artists.
3			record_label	VARCHAR(255)	Yes	CD record label.
4			tracks	JSONB	Yes	Track list. Each track may include title and length.
5			genre	VARCHAR(100)	Yes	CD genre.
6			release_date	DATE	No	CD release date.

Constraints:

- PRIMARY KEY (product_id)
- FOREIGN KEY (product_id) REFERENCES products(product_id) ON DELETE CASCADE
- artists and tracks are stored as JSONB to keep the physical schema consistent with the simplified final ERD.

dvds

#	PK	FK	Column name	Data type	Mandatory	Description
1	x	x	product_id	UUID	Yes	ID of the corresponding product
2			disc_type	VARCHAR(50)	Yes	DVD disc type: BLU_RAY or HD_DVD.
3			director	VARCHAR(255)	Yes	DVD director
4			runtime	INTEGER	Yes	DVD runtime
5			studio	VARCHAR(255)	Yes	DVD studio
6			language	VARCHAR(100)	Yes	DVD language
7			subtitles	JSONB	Yes	DVD subtitles
8			release_date	DATE	No	DVD release date
9			genre	VARCHAR(100)	No	DVD genre

Constraints:

- PRIMARY KEY (product_id)
- FOREIGN KEY (product_id) REFERENCES products(product_id) ON DELETE CASCADE
- CHECK (runtime > 0)
- CHECK (disc_type IN ('BLU_RAY', 'HD_DVD'))
- subtitles is stored as JSONB to support multiple subtitle values

newspapers

#	PK	FK	Column name	Data type	Mandatory	Description
1	x	x	product_id	UUID	Yes	ID of the corresponding product
2			editor_in_chief	VARCHAR(255)	Yes	Newspaper editor-in-chief

3			publisher	VARCHAR(255)	Yes	Newspaper publisher
4			publication_date	DATE	Yes	Newspaper publication date
5			issue_number	VARCHAR(100)	No	Newspaper issue number
6			publication_frequency	VARCHAR(100)	No	Publication frequency
7			issn	VARCHAR(100)	No	ISSN code
8			language	VARCHAR(100)	No	Newspaper language
9			sections	JSONB	No	Newspaper sections, such as politics, business, sports, or culture.

Constraints:

- PRIMARY KEY (product_id)
- FOREIGN KEY (product_id) REFERENCES products(product_id) ON DELETE CASCADE
- sections is stored as JSONB because sections are treated as multivalued product attributes

product_histories

#	PK	FK	Column name	Data type	Mandatory	Description
1	x		history_id	UUID	Yes	Unique identifier of a product history record.
2		x	product_id	UUID	Yes	Product affected by the operation
3		x	performed_by	UUID	Yes	User who performed the operation
4			action_type	history_action_type_enum	Yes	Type of action: CREATE, UPDATE, DELETE, DEACTIVATE, or STOCK_ADJUST.
5			action_time	TIMESTAMP	Yes	Time when the action was performed
6			old_value_snapshot	JSONB	No	Snapshot of product data before the operation
7			new_value_snapshot	JSONB	No	Snapshot of product data after the operation.

8			reason	TEXT	No	Additional note; also used to record stock adjustment reason if applicable.
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Constraints:

- PRIMARY KEY (history_id)
- FOREIGN KEY (product_id) REFERENCES products(product_id) ON DELETE CASCADE
- FOREIGN KEY (performed_by) REFERENCES users(user_id) ON DELETE RESTRICT
- action_type must be one of CREATE, UPDATE, DELETE, DEACTIVATE, STOCK_ADJUST

orders

#	PK	FK	Column name	Data type	Mandatory	Description
1	x		order_id	UUID	Yes	Unique identifier of an order
2		x	processed_by	UUID	No	Product manager who processes, approves, or rejects the order.
3			status	order_status_enum	Yes	Current order status
4			total_amount	NUMERIC(14,2)	Yes	Total amount of the order
5			order_view_token	UUID	Yes	Token used in customer order-view link
6			cancel_token	UUID	Yes	Token used in customer order-cancellation link
7			processed_at	TIMESTAMP	No	Time when the order was processed by a product manager
8			created_at	TIMESTAMP	Yes	Order creation timestamp
9			updated_at	TIMESTAMP	Yes	Last order update timestamp

Constraints:

- PRIMARY KEY (order_id)

- FOREIGN KEY (processed_by) REFERENCES users(user_id) ON DELETE SET NULL
- CHECK (total_amount >= 0)
- UNIQUE (order_view_token)
- UNIQUE (cancel_token)
- status must be one of PENDING_PROCESSING, APPROVED, REJECTED, CANCELLED.

order_items

#	PK	FK	Column name	Data type	Mandatory	Description
1	x		order_item_id	UUID	Yes	Unique identifier of an order item
2		x	order_id	UUID	Yes	Order containing this item
3		x	product_id	UUID	Yes	Product referenced by this order item
4			product_title	VARCHAR(255)	Yes	Product title snapshot at ordering time
5			unit_price	NUMERIC(14,2)	Yes	Product unit price snapshot at ordering time
6			quantity	INTEGER	Yes	Quantity ordered
7			line_amount_excl_vat	NUMERIC(14,2)	Yes	Line amount excluding VAT
8			line_amount_incl_vat	NUMERIC(14,2)	Yes	Line amount including VAT

Constraints:

- PRIMARY KEY (order_item_id)
- FOREIGN KEY (order_id) REFERENCES orders(order_id) ON DELETE CASCADE
- FOREIGN KEY (product_id) REFERENCES products(product_id) ON DELETE RESTRICT
- CHECK (unit_price >= 0)
- CHECK (quantity > 0)

- CHECK (line_amount_excl_vat >= 0)
- CHECK (line_amount_incl_vat >= 0)
- UNIQUE (order_id, product_id)

delivery_infos

#	PK	FK	Column name	Data type	Mandatory	Description
1	x		delivery_info_id	UUID	Yes	Unique identifier of delivery information
2		x	order_id	UUID	Yes	Order associated with this delivery information
3			customer_name	VARCHAR(255)	Yes	Customer name
4			phone_number	VARCHAR(30)	Yes	Customer phone number
5			email	VARCHAR(255)	Yes	Customer email
6			delivery_address	TEXT	Yes	Delivery address
7			delivery_province	VARCHAR(100)	Yes	Delivery province or city
8			delivery_instructions	TEXT	No	Optional delivery instructions

9			expected_date	DATE	No	Expected delivery date
10			shipping_fee	NUMERIC(14,2)	Yes	Shipping fee

Constraints:

- PRIMARY KEY (delivery_info_id)
- FOREIGN KEY (order_id) REFERENCES orders(order_id) ON DELETE CASCADE
- UNIQUE (order_id)
- CHECK (shipping_fee >= 0)

invoices

#	PK	FK	Column name	Data type	Mandatory	Description
1	x		invoice_id	UUID	Yes	Unique identifier of an invoice
2		x	order_id	UUID	Yes	Order associated with this invoice
3			total_product_price_excl_vat	NUMERIC(14,2)	Yes	Total product price excluding VAT
4			vat_amount	NUMERIC(14,2)	Yes	VAT amount
5			total_product_price_incl_vat	NUMERIC(14,2)	Yes	Total product price including VAT
6			delivery_fee	NUMERIC(14,2)	Yes	Delivery fee
7			total_amount_to_pay	NUMERIC(14,2)	Yes	Final amount to be paid by the customer

8			created_at	TIMESTAMP	Yes	Invoice creation timestamp
9			updated_at	TIMESTAMP	Yes	Last invoice update timestamp

Constraints:

- PRIMARY KEY (invoice_id)
- FOREIGN KEY (order_id) REFERENCES orders(order_id) ON DELETE CASCADE
- UNIQUE (order_id)
- CHECK (total_product_price_excl_vat >= 0)
- CHECK (vat_amount >= 0)
- CHECK (total_product_price_incl_vat >= 0)
- CHECK (delivery_fee >= 0)
- CHECK (total_amount_to_pay >= 0)
- CHECK (total_product_price_incl_vat = total_product_price_excl_vat + vat_amount)
- CHECK (total_amount_to_pay = total_product_price_incl_vat + delivery_fee)

payment_transactions

#	PK	FK	Column name	Data type	Mandatory	Description
1	x		transaction_id	UUID	Yes	Unique identifier of a payment transaction
2		x	invoice_id	UUID	Yes	Invoice paid by this payment transaction
3			transaction_content	TEXT	No	Payment transaction content or message
4			transaction_datetime	TIMESTAMP	Yes	Time of the payment transaction
5			amount	NUMERIC(14,2)	Yes	Paid amount
6			status	payment_status_enum	Yes	Payment status
7			payment_method	payment_method_enum	Yes	Payment method: QR_CODE or CREDIT_CARD
8			error_code	VARCHAR(100)	No	Error code returned by payment gateway if payment fails

9			gateway_transaction_ref	VARCHAR(255)	No	Transaction reference returned by the payment gateway
10			created_at	TIMESTAMP	Yes	Record creation timestamp

Constraints:

- PRIMARY KEY (transaction_id)
- FOREIGN KEY (invoice_id) REFERENCES invoices(invoice_id) ON DELETE CASCADE
- CHECK (amount >= 0)
- status must be one of PENDING, SUCCESS, FAILED, CANCELLED, REFUNDED.
- payment_method must be one of QR_CODE, CREDIT_CARD

refund_transactions

#	PK	FK	Column name	Data type	Mandatory	Description
1	x		refund_id	UUID	Yes	Unique identifier of a refund transaction
2		x	payment_transaction_id	UUID	Yes	Payment transaction being refunded.
3			refund_amount	NUMERIC(14,2)	Yes	Refund amount
4			refund_reason	TEXT	Yes	Reason for refund
5			refund_datetime	TIMESTAMP	Yes	Time of refund transaction
6			refund_status	refund_status_enum	Yes	Refund status
7			refund_method	refund_method_enum	Yes	Refund method
8			manual_refund_note	TEXT	No	Required note when refund is handled manually
9			created_at	TIMESTAMP	Yes	Record creation timestamp

Constraints:

- PRIMARY KEY (refund_id)

- FOREIGN KEY (payment_transaction_id) REFERENCES payment_transactions(transaction_id) ON DELETE CASCADE
- CHECK (refund_amount >= 0)
- CHECK (refund_method <> 'MANUAL_BANK_TRANSFER' OR manual_refund_note IS NOT NULL)
- refund_status must be one of PENDING, SUCCESS, FAILED, MANUAL_REQUIRED.
- refund_method must be one of PAYPAL_API, MANUAL_BANK_TRANSFER